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|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of the Student</b> | <b>AKHILAPOCHIMIREDDY</b>                                                                                                                                                          |
| <b>Reg. No</b>             | <b>192425225</b>                                                                                                                                                                   |
| <b>GitHub Link</b>         | <b><a href="https://github.com/120608akhila/MLA0405-DEEP-LEARNING/tree/main/Assessment/CO5">https://github.com/120608akhila/MLA0405-DEEP-LEARNING/tree/main/Assessment/CO5</a></b> |

## SIMATS ENGINEERING

### CO5 - ASSESSMENT TOOL 3

#### Technical Presentation

**COURSE NAME:** Deep Learning  
**SLOT :**

**COURSE CODE:** MLA04  
**Faculty :** Dr.Arul Raj A.M

#### COURSE OUTCOME COVERED

**CO5:** Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

**Course Outcome Weightage: 35%**  
**Assessment Tool Weightage: 50%**

**Duration: 90 minutes**  
**Mark : 25**

#### Code of Conduct:

I **AKHILA POCHIMIREDDY(192425225)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:** Akhila.P **Name of the student:** AKHILA POCHIMIREDDY

| Criteria         | Understanding of Problem Context (2) | Application of Concepts (3) | Problem-Solving Approach (3) | Accuracy of Solution (2) | Clarity (1) | Total (10)  |
|------------------|--------------------------------------|-----------------------------|------------------------------|--------------------------|-------------|-------------|
| <b>Weightage</b> | <b>20 %</b>                          | <b>30%</b>                  | <b>30%</b>                   | <b>20 %</b>              | <b>10 %</b> | <b>100%</b> |
| <b>Q1</b>        |                                      |                             |                              |                          |             |             |
| <b>Total</b>     |                                      |                             |                              |                          |             |             |

**Signature of the Faculty:**

**Total Marks:**

**PHOTO :**



**QUESTIONS:**

### **3. Deep Belief Networks and Boltzmann Machines**

Develop a technical presentation comparing Deep Belief Networks and Boltzmann Machines. Explain their architecture, energy-based learning, hidden and visible units, training process, and feature-learning capabilities. Illustrate their use in a suitable real-world application.

**PRESENTATION ANSWER :**

| Topic            | Boltzmann Machine (BM)                    | Deep Belief Network (DBN)                |
|------------------|-------------------------------------------|------------------------------------------|
| Architecture     | Visible and hidden units with connections | Multiple stacked hidden layers           |
| Units            | Visible + hidden                          | Visible + multiple hidden layers         |
| Learning         | Energy-based learning                     | Layer-wise unsupervised learning         |
| Training         | Contrastive Divergence                    | Greedy layer-wise training               |
| Feature Learning | Learns latent features                    | Learns hierarchical features             |
| Main Advantage   | Models complex probability distributions  | Learns deep representations efficiently  |
| Application      | Pattern recognition                       | Image recognition and feature extraction |

**Working**

**BM:** Input → Visible Units → Hidden Units → Energy Calculation → Weight Update.

**DBN:** Input → Hidden Layer 1 → Hidden Layer 2 → Hidden Layer 3 → Learned Features.

### Real-World Application

**Image recognition:** DBNs can learn hierarchical image features such as edges, shapes, and objects for classification.

### RUBRICS FOR CALCULATION:

| Criteria                         | Weightage | Level 4 Exemplary                                                          | Level 3 Proficient                                    | Level 2 Developing                              | Level 1 Beginning                                      |
|----------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding ; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy    | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies        | Illogical or unclear approach                          |
| Accuracy of Solution             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                      | Incorrect or incomplete solution                       |
| Clarity                          | 10        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                 | Poorly presented and difficult to understand           |